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SUPERSEDING
MIL-J-38229B(USAF)
1 June 1972

MILITARY SPECIFICATION

JACK, HYDRAULIC, TRIPOD MMU-59/E

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 20-ton, hydraulic tripod jack, designated MMU-59/E.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, and Seals
PPP-B-621	Box, Wood, Nailed and Lock Corner

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, And Ordnance
MIL-H-81019	Hydraulic Fluid, Petroleum Base,Ultra-Low Temperature

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-808	Finish, Materials, and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWING

AIR FORCE

64J35484	Jack Hydraulic Tripod, 20 Ton Capacity, MMU-59/E.
7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-3951-82 Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. This specification makes provisions for first article testing. (See 4.2 and 6.3)

3.2 General requirements. The requirements of Drawing 64J35484 apply as requirements of this specification. When the drawing and this specification conflict, this specification shall govern.

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3.3 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.4 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions and marking of parts and assemblies.

3.5 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-808 in conjunction with AF drawing 7545352.

3.6 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. The inspection and testing of the jack shall be classified as follows:

a.	First article	See 4.4
b.	Acceptance tests	See 4.5

4.3 Test conditions. The jack shall be serviced with hydraulic fluid conforming to MIL-H-5606 except for the low temperature test which shall be conducted with fluid conforming to MIL-H-81019.

4.4 First article (see 6.3).

4.4.1 Test sample. One jack shall be subjected to the tests specified in 4.4.2.

4.4.2 First article tests. The first article tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests (4.5.1)
b. Sampling plan and tests (4.5.2)

4.5.1 Individual tests. Each jack shall be subjected to the following tests as described under 4.6:

a. Examination of product (4.6.1)
b. Ram and cylinder hydrostatic pressure (4.6.2)
c. Pump bypass relief valve test (4.6.3)

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4.5.2 Sampling plan and tests.

4.5.2.1 Periodic sampling. Five jacks shall be selected at random from every fifty or fraction thereof produced and subjected to the static proof test as described under 4.6.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawing.

4.6.2 Ram and cylinder hydrostatic pressure. The ram and cylinder assembly shall be proof tested at 3,000 pounds per square inch (psi) hydraulic pressure. Any leakage of hydraulic fluid shall be cause for rejection.

4.6.3 Pump bypass relief valve. The pump shall be proof-tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted as specified on the drawing.

4.6.4 Life test. With the screw extension retracted, the jack shall be subjected to the following tests.

4.6.4.1 Vertical lift. The vertical lift test shall consist of 200 cycles of raising and lowering a vertical load of 20 tons. One cycle shall consist of raising the hydraulic ram under load from the fully retracted position to the fully extended position and return to the fully retracted position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.4.2 Dynamic side load. The dynamic side load test shall consist of 60 cycles of raising a 20-ton load vertically with a graduated 3 ton horizontal load. Twenty cycles each shall be applied through the jack pad socket adapter in the direction of an opposite jack leg. The horizontal load on the jack adapter shall be gradually increased from zero in the fully retracted position to 3 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5 Static proof. After the bypass relief valve has been adjusted for overload, the jack shall be tested with the screw extension in the fully

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extended position by raising a 30 ton load vertically from the lowest position on the hydraulic ram to the fully extended position. While in the extended position, the vertical load shall be reduced to 20 tons and maintained during application of a 3 ton horizontal load applied to the extended screw extension. To complete the cycle, the horizontal and vertical loads shall be removed and the jack ram allowed to retract to its original position with no more than 50 pounds applied to the top of the ram. Failure of any part of the jack during this test shall be cause for rejection.

4.6.6 Low temperature. The jack shall be subjected to low temperature in accordance with MIL-STD-810, method 502, procedure 1. A temperature of -54° Centigrade (C) shall be maintained for a period of 24 hours. At the end of the exposure period and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the by-pass relief valve operates. The lowering valve shall be opened, and the ram shall retract with not more than 50 pounds applied to the ram. For this test, the jack shall be serviced with hydraulic fluid conforming to MIL-H-81019. The time and number of full strokes necessary to fully extend the ram shall be recorded at both $+21^{\circ}$ and -54°C. The time and strokes required to fully extend the ram at -54°C shall be not more than double the time and strokes required at $+21^{\circ}$C.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1. Preservative Applications. Preservatives in accordance with MIL-P-116 shall be used.

5.1.1.2 Preservation. Unless otherwise specified (see 6.2), each Hydraulic Jack shall be individually preserved in accordance with Method I of MIL-P-116 designated herein insuring compliance with the applicable requirements of that specification. Each jack shall be in a disassembled condition with jack fully retracted. The tow bar shall be secured in an upright position to the jack leg.

5.1.2 Level C. The level C preservation for the Hydraulic Jack shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial preservation for the Hydraulic Jack shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2).

5.2.1 Level A. The packaged Jack shall be packed in containers conforming to PPP-B-621, Class 2, Overseas. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I or IV, Finish A. When gross weight exceeds 200 pounds or

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the container length and width is 48" X 24" or more and the weight exceeds 100 pounds, 3x4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.2.2 Level B. The packaged Jack shall be packed in containers conforming to PPP-B-621 Class 1, Domestic. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I, or IV, finish A. When gross weight exceeds 200 pounds or the container length and width is 48" X 24" or more and the weight exceeds 100 pounds 3X4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.2.3 Level C. The Level C packing for this jack shall conform to the MIL-STD-794 requirements of the container specification.

5.2.4 Commercial. The preserved jack shall be packed in accordance with the requirements of ASTM D 3951.

5.2.5 Palletized Loads. When specified (see 6.2) Palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D 3951.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3 and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.1.

5.4.3 Physical Protection. Interior blocking, bracing, cushioning, et cetera, shall be in accordance with MIL-STD-1186.

6. NOTES

6.1 Intended use. The MMU-59/E jack is intended for use as a wing jack for aircraft whose jack pads are between 36 and 152 inches from the ground.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Location and conditions for first article testing (see 4.4).

c. Selection and applicable level of preservation, packaging, and packing (see 5.1 and 5.2).

d. When palletized loads are required (see 5.2.5).

* 6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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